

Experiment No.1a

Title : How to use Keil u Vision 4 for ARM7- LPC2148

Aim : To learn Keil u Vision 4 IDE for ARM7- LPC2148

Software: Keil IDE, Flash Magic

Theory:

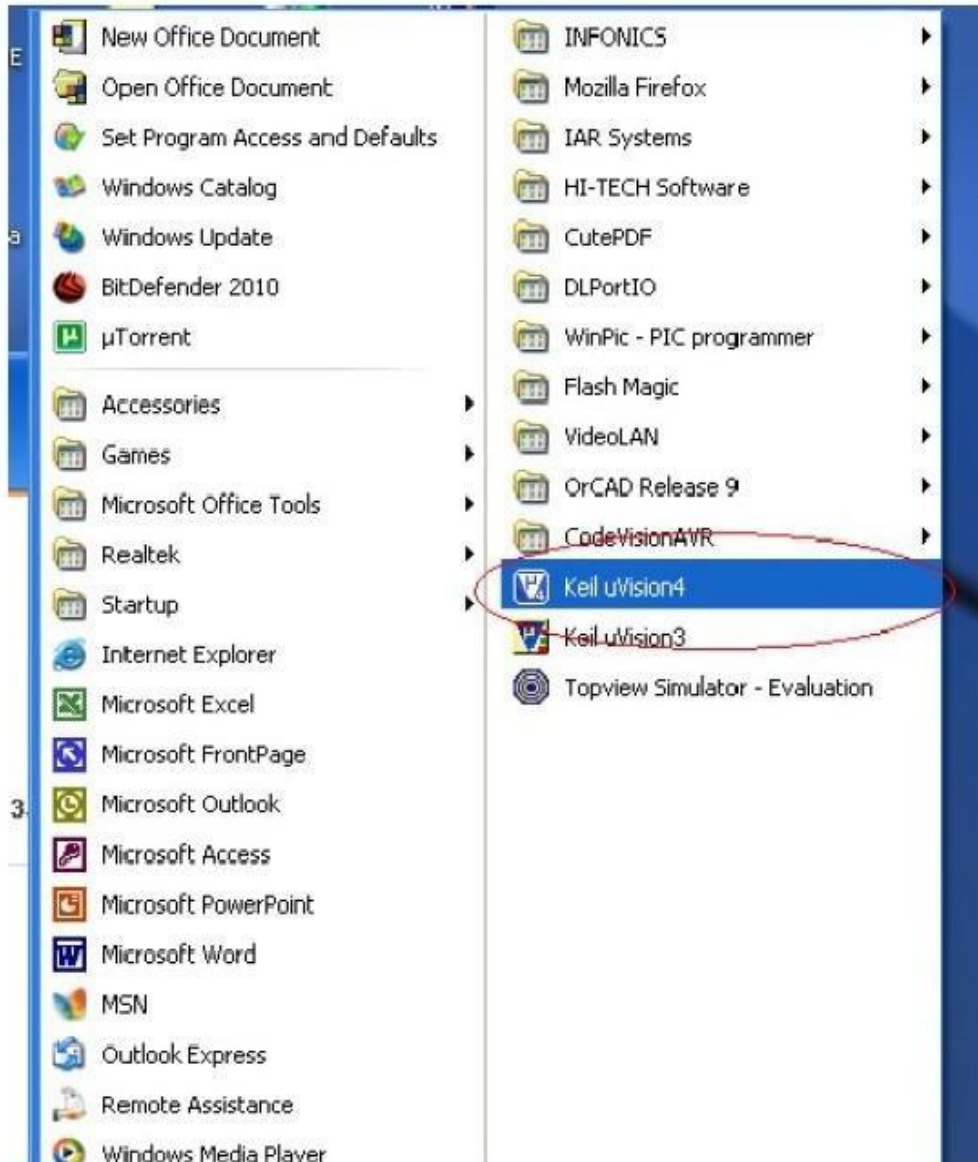
HOW TO USE KEIL μ VISION4

For ARM7 (LPC2148)
Step By Step

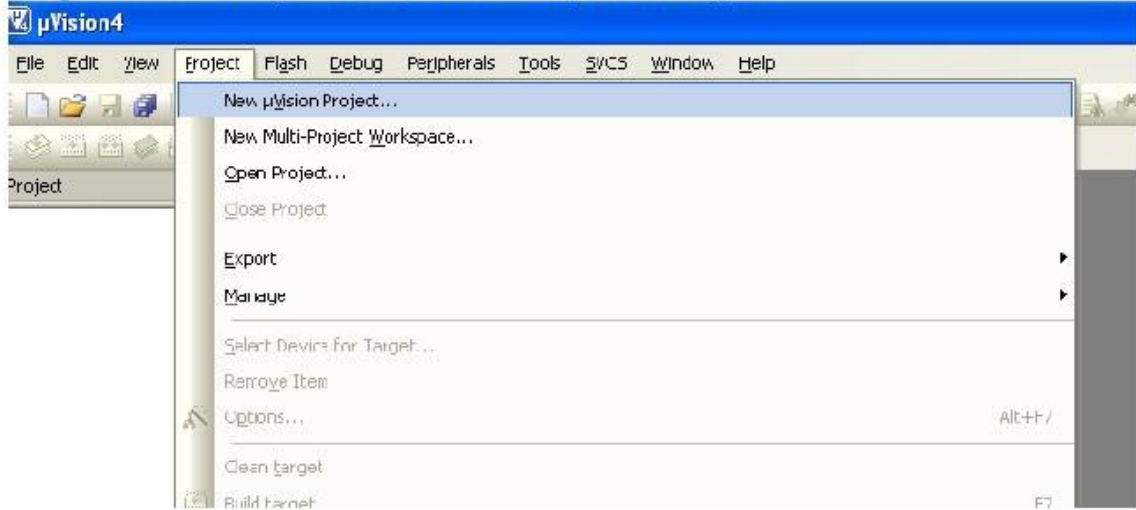
For Running Code in Flash Memory



Step1: Click for **KEIL μ VISION4** Icon . Which appearing after Installing Keil KEIL μ VISION4.



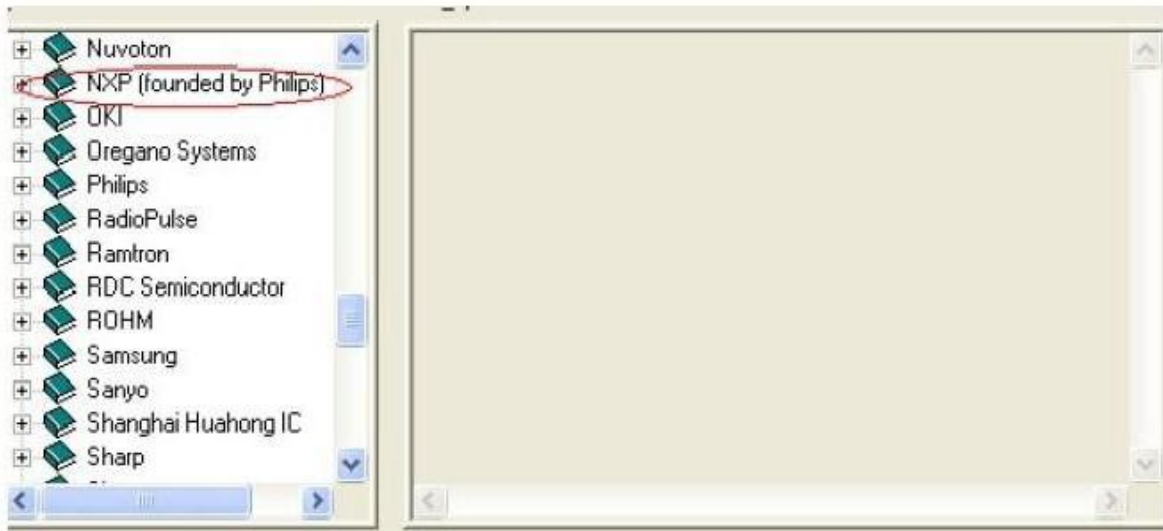
Step2: Click on Project Menu, Then **New μ Vision Project**.



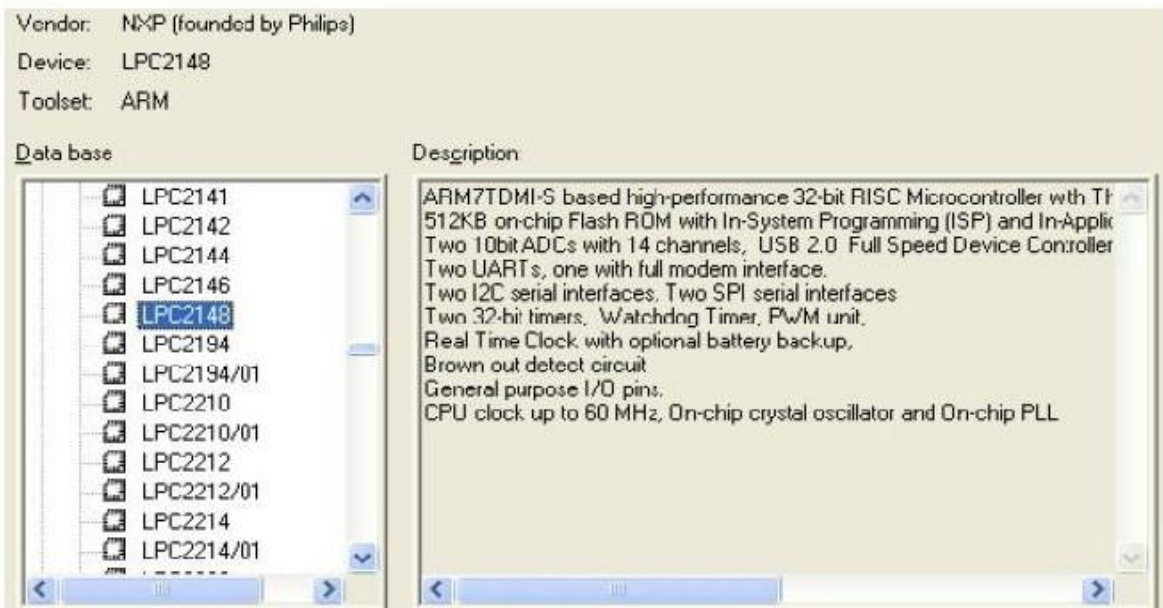
Step3: Create New Project Folder named as “**Keil Test**”.



Step 4: Select Target Device Vendor (i.e.).



Step 5: Then select specific chip LPC2148.



Step 6: Then select specific chip i.e. LPC2148.



Step 7: Then you will see following window.



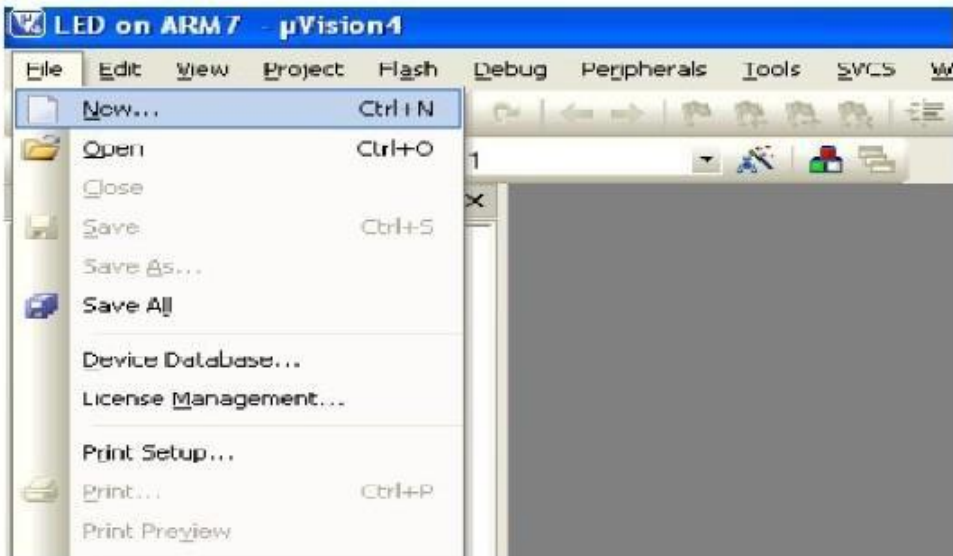
Step 8: Now you see Startup.s is already added which is necessary for running code for Keil.

Note: Code will Not Run without Startup.s
Startup.s is available in C:\Keil\ARM\Startup\Philips.

The startup-code executes immediately upon reset of the target system and performs the following operations:

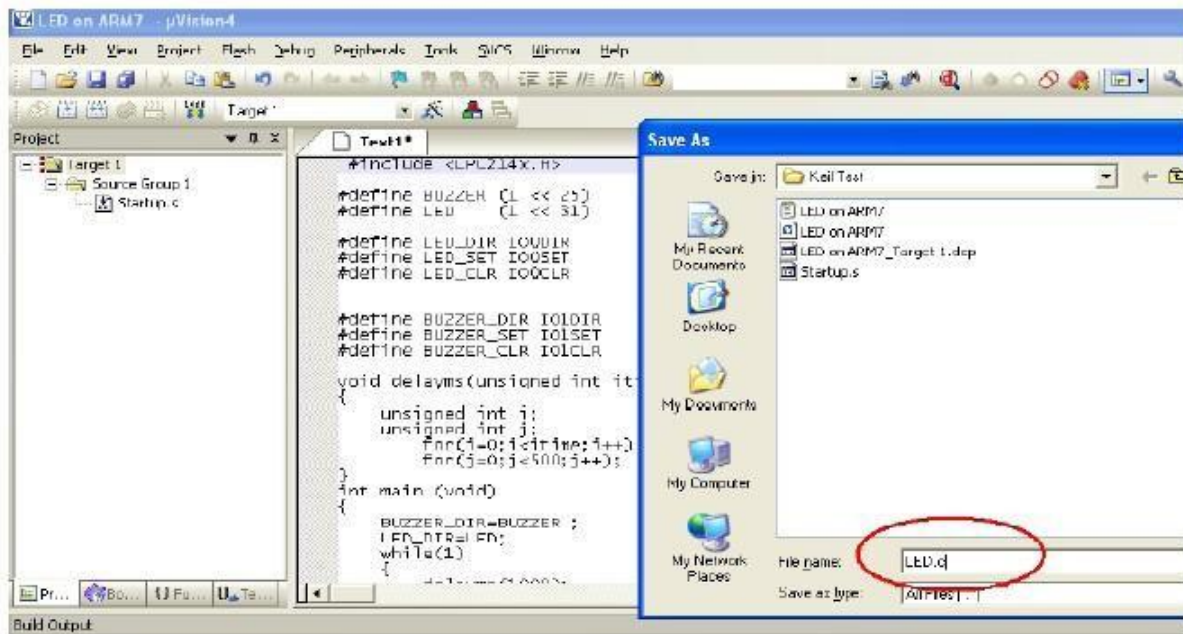
1. Defines interrupt and exception vectors.
2. Configures the CPU clock source (on some devices).
3. Initializes the external bus controller.
4. Copies the exception vectors from ROM to RAM for systems with memory remapping.
5. Initializes other low level peripherals, if necessary.
6. Reserves and initializes the stack for all modes.
7. Reserves the heap.
8. Transfers control to the main C function.

Step 9: Now Click on File Menu and Click on New.

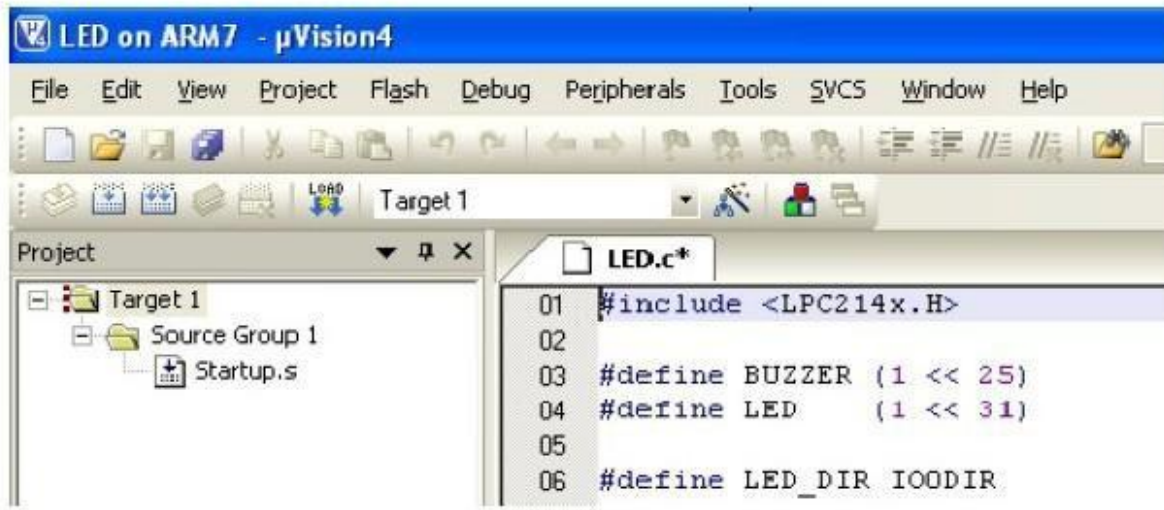


Step 10: Write Code for Blink LED in C and **FileName.c** Save.

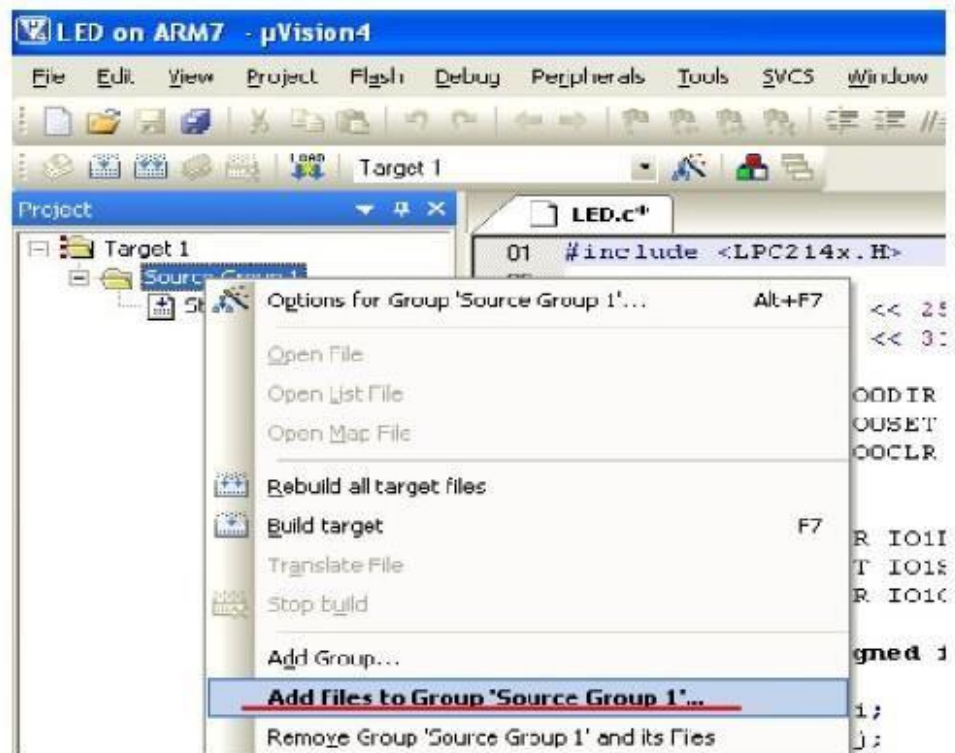
Note: Don't forget to save **.c Extension**.



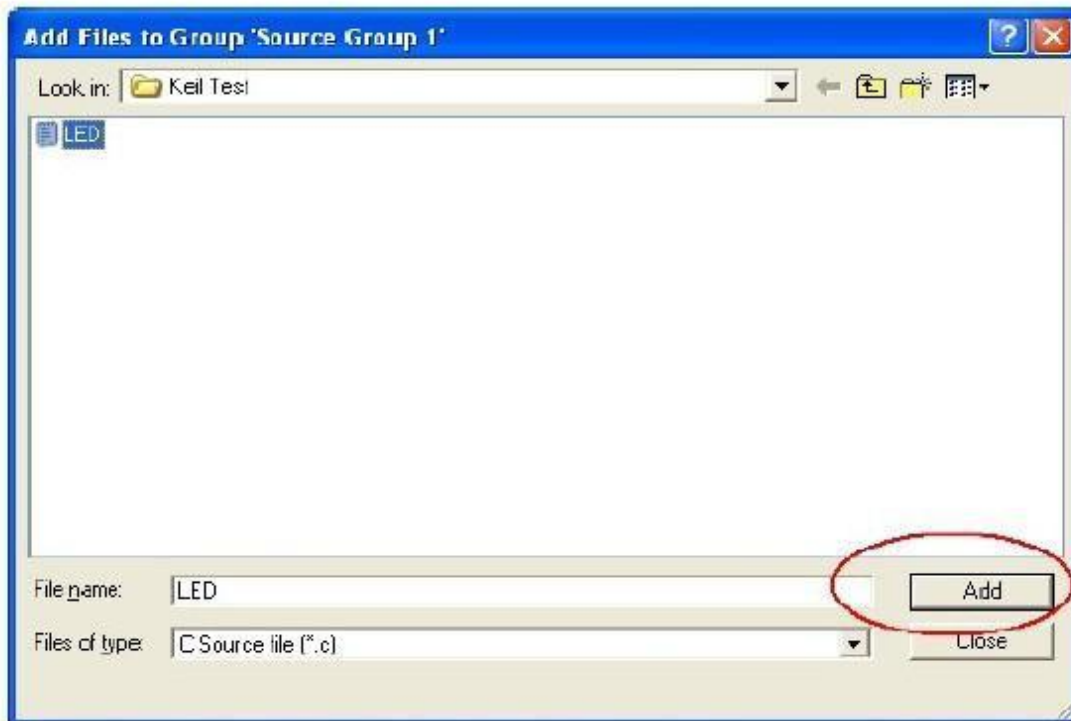
Step 11: Now you Window in C Syntax.



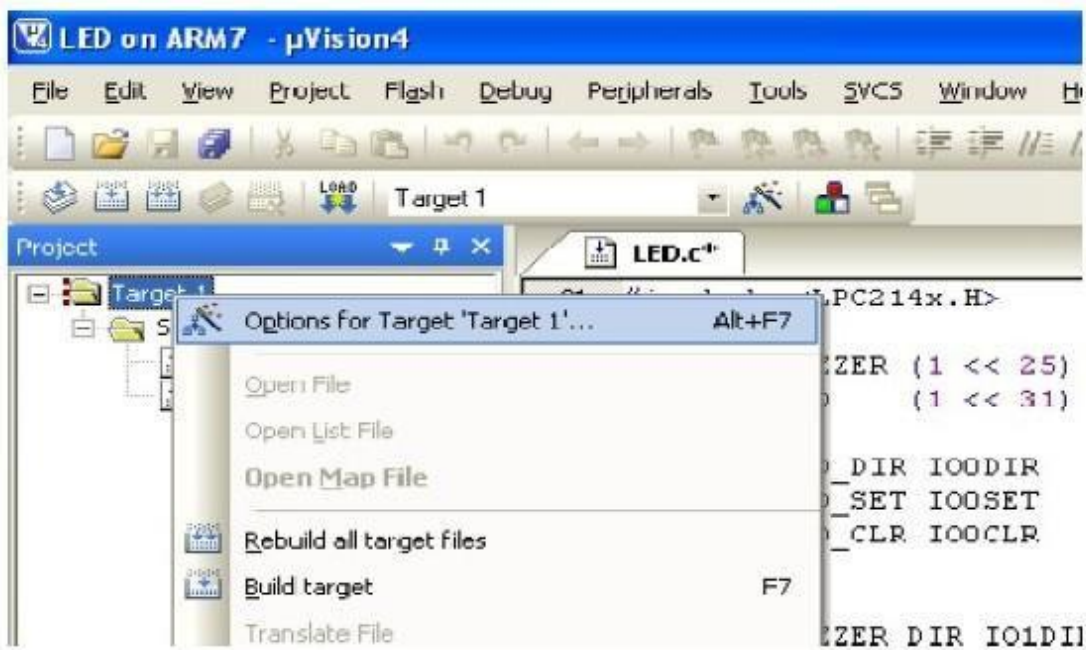
Step 12: Now you add LED.c file by adding Source Group 1 Add files to Group 'Source Group 1'.



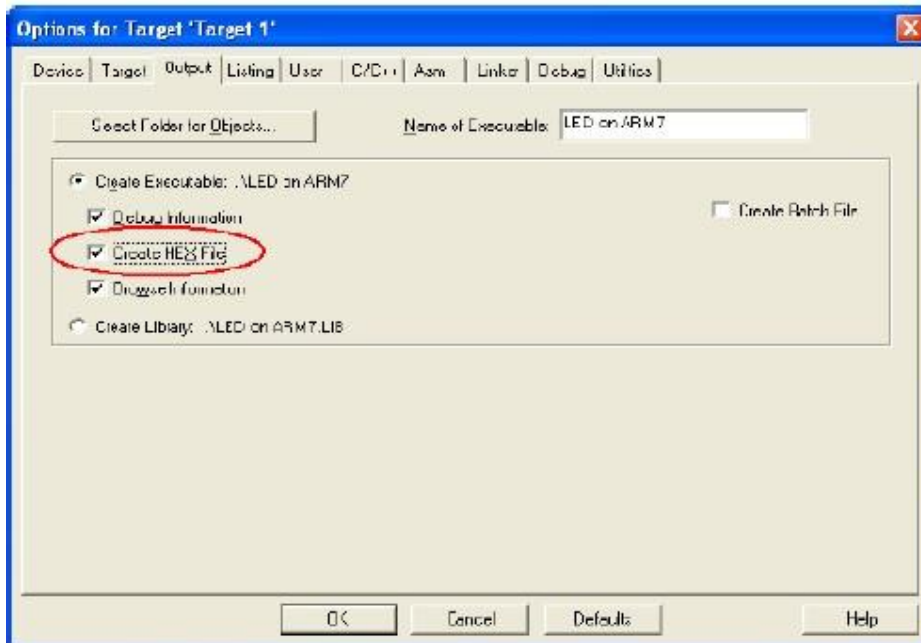
Step 13: Add LED.C file.



Step 14: Now Click on Options for Target 'Target 1'.



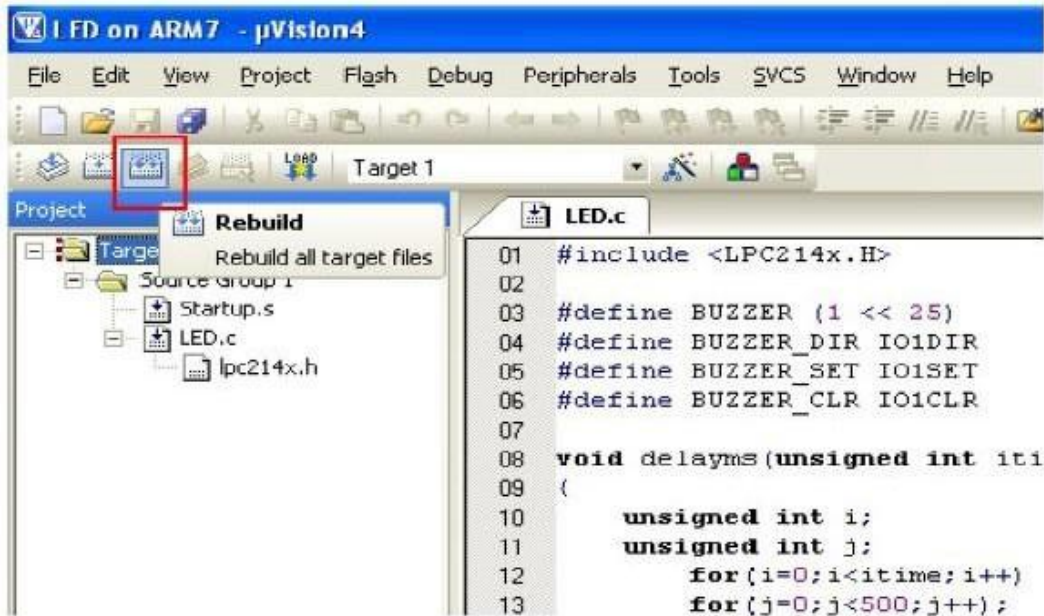
Step 15: Go to Options for Target 'Target 1'. Click on Check Box Create HEX File.



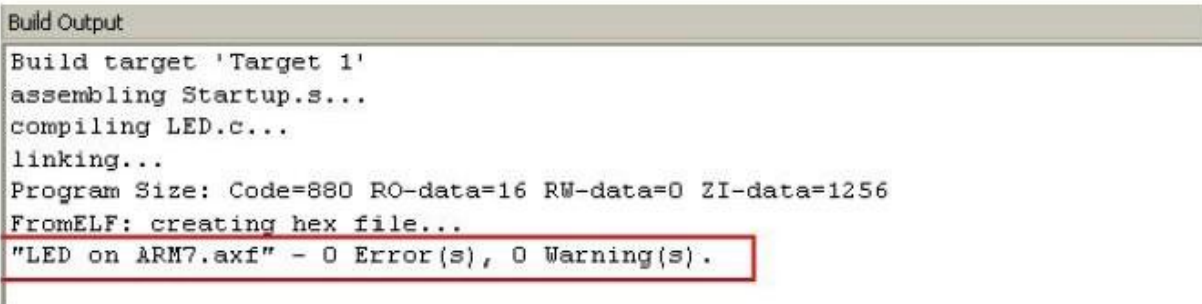
Step 16: Then go to Linker. Click on Use Memory Layout for Target Dialog.



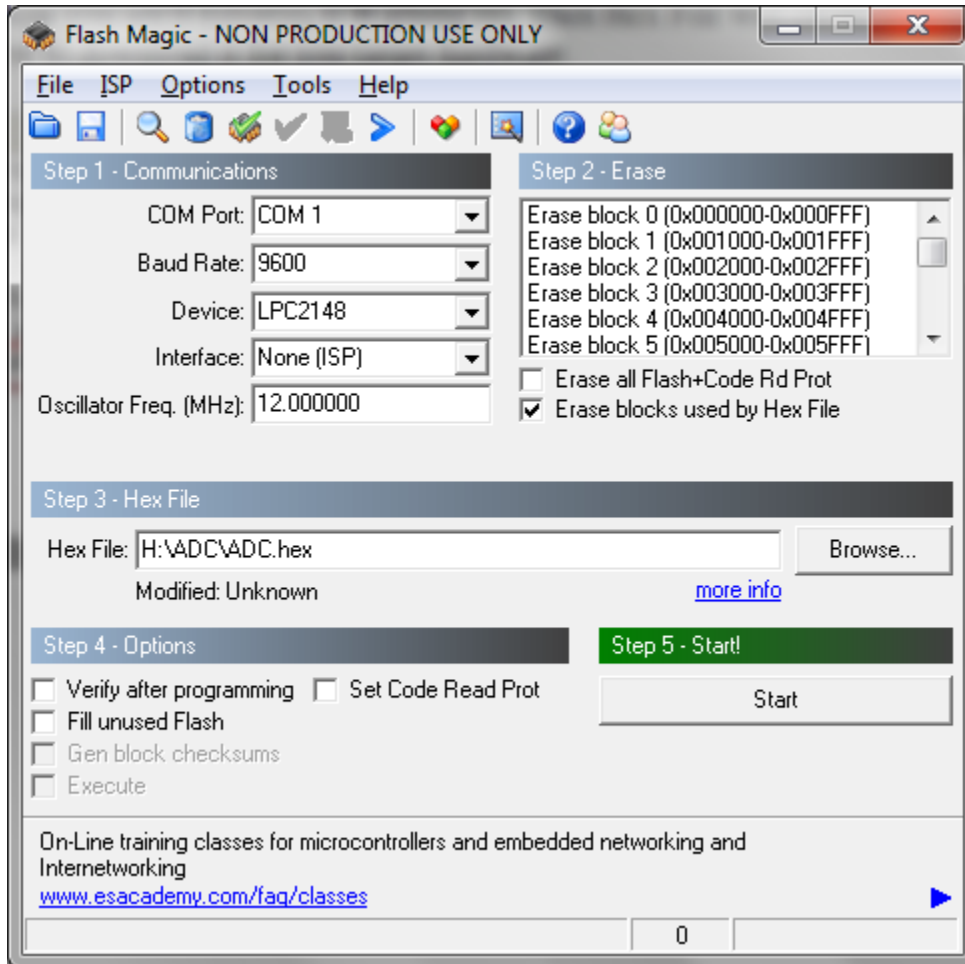
Step 16: Then Click on Rebuild All Target Files



Step 17: Now you see 0 Error(s), 0 Warning (s). Then Hex File will create in Specific Folder. Now to download it for you target hardware.



Downloading code (HEX file) to LPC2148 board



1. Open Flash magic software.
2. Select the appropriate COM port.
3. Select Baud rate as 9600 or 38400.
4. Select Device as LPC2148.
5. Keep Interface as None (ISP).
6. Make Oscillator Freq as 12.00.
7. Check in Erase blocks used by Hex File
8. Browse and select particular hex file to be downloaded on the board.
9. Check the switch position on board & make it towards ISP & press reset.

Advanced Processor

10. Click on Start to start downloading.
11. On board make switch position towards RUN ,press reset & verify the result.